

definition: 1. PDF (Probability Density Function), 概率密度函数.

2. CDF (Cumulative Distribution Function), 累积分布函数.

$PDF_R(r) = Pr[R=r]$, R 恰好等于某个值的概率.

$CDF_R(r) = Pr[R \leq r]$, R 小于或等于某个值的概率.

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⇒ KE! 随机变量只是整理了一下样本空间, 将它划分为不同的区块, 每个区块中的不同元素 (即区块中的结果) 在这个随机变量下都有相同的值.

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⇒ 1. 函数无区别: Both PDF_R and CDF_R capture the same information about R , so take your choice. The key point here is that neither the probability density function nor the cumulative distribution function involves the sample space of an experiment.

$R: S \rightarrow T$, 都会根据函数相同这个等价关系, 将定义域 S 划分为互不相交的子集, 每个子集里的元素在 R 下具有相同的值. One of the really interesting things about density functions and distribution functions is that many random variables turn out to have the *same* pdf and cdf. In other words, even though R and S are different random variables on different probability spaces, it is often the case that

$$PDF_R = PDF_S.$$

In fact, some pdf's are so common that they are given special names. For example, the three most important distributions in computer science are the *Bernoulli distribution*, the *uniform distribution*, and the *binomial distribution*. We look more closely at these common distributions in the next several sections.

2. 划分对应等价关系

$$\omega_1 \sim \omega_2 \Leftrightarrow R(\omega_1) = R(\omega_2)$$

⇒ 它不只是一个函数, 更是一个

分类器.

即 KE

其用处在于逻辑抽象和降维计算.

eg. 1% 的故障率, 一车该换零件?

令 $R = \begin{cases} 1, & \text{故障} \\ 0, & \text{将故障} \end{cases}$

for some $p \in [0, 1]$. The corresponding cumulative distribution function is $F_p: \mathbb{R} \rightarrow [0, 1]$ where

$$f_p(0) = p, \text{ and}$$

$$f_p(1) = 1 - p,$$

$$E[R] = 1 \times 1\% + 0 \times (1 - 1\%)$$

$$F_p(x) ::= \begin{cases} 0 & \text{if } x < 0 \\ p & \text{if } 0 \leq x < 1 \\ 1 & \text{if } 1 \leq x. \end{cases}$$

⇒ $E[X] = E[R_1 + R_2 + R_3 + \dots]$

18.3.2 Uniform Distributions 均匀分布.

$$= E[R_1] + E[R_2] + \dots$$

A random variable that takes on each possible value in its codomain with the same probability is said to be *uniform*. If the codomain V has n elements, then the *uniform distribution* has a pdf of the form

$$= 0.1\% + 0.1\% + \dots$$

$$PDF \quad f: V \rightarrow [0, 1]$$

$$= 1.$$

where

$$f(v) = \frac{1}{n}$$

⇒ 用于将逻辑化为算术, 将

离散故障的局部.

for all $v \in V$.